Updated April 8, 2026

Pietro Visaggio

visaggip@bc.edu • pietrovisaggio.com • Research • CV (latest) • LinkedIn

Ph.D. candidate specializing in Industrial Organization and Energy Economics with expertise in electricity markets, battery storage, and renewable technologies. My research investigates how Battery Energy Storage Systems (BESS) are integrated into electricity markets, and how their utilization shapes outcomes such as electricity prices, consumer welfare, and the effective integration of renewable resources.

Fields of Interest

Industrial Organization, Energy Economics

Education

Ph.D. in Economics, **Boston College**	May 2026 (Expected)
M.A. in Economics, **Boston College**	2022
M.Sc. in Economics and Finance, **LUISS & EIEF**	2020
B.A. in Economics and Finance, **Università di Tor Vergata**	2018

Working Papers

"Pairing Batteries with Renewables: How Ownership Shapes Operational Incentives and Market Outcomes"

This paper examines how battery storage ownership structure affects wholesale electricity market outcomes by shaping operational incentives. Using a dynamic dispatch model calibrated to Texas data, I show how transmission congestion creates conditions in which batteries operated jointly with a renewable plant are used strategically to increase the value of renewable production. The strength of this incentive depends on supply elasticity and the timing of renewable production. Co-owned batteries earn roughly 76 percent higher profits than standalone batteries in markets where strategic incentives arise. Despite this strategic behavior, co-owned and standalone batteries produce similar effects on consumer surplus, renewable curtailment, and carbon emissions. While market conditions do not generate enough profits for battery investment to be viable—regardless of ownership—the positive effects on consumer surplus and carbon emissions make batteries desirable from consumers' perspective. Under a uniform subsidy policy, co-ownership's higher profitability makes more batteries viable at moderate subsidy rates.

Work in Progress

"Estimating the Curtailment-Mitigating Role of Battery Energy Storage Systems"

The rapid expansion of variable renewable energy (VRE) in Texas—reaching 27 GW of solar and 43 GW of wind capacity by 2024—has been accompanied by rising curtailment. When available generation exceeds transmission capacity or contemporaneous demand, grid operators must curtail zero-marginal-cost renewable output. This paper quantifies how battery energy storage systems (BESS) mitigate curtailment by absorbing surplus generation. Using hourly Texas data from 2019-2024, I exploit the staggered deployment of new battery installations to estimate the causal effect of storage on market-level curtailment. The identification strategy relies on exogenous variation in battery deployment driven by declining capital costs. I find that each additional MWh of battery capacity reduces curtailment by approximately 0.1 MWh during daytime (9 AM to 6 PM), when solar generation is abundant, but has negligible effects during nighttime when wind production peaks.

Conferences and Seminars

ISO New England Market Design Workshop (September 2025)

Berkeley/Sloan Summer School in Environmental and Energy Economics (August 2024)

Teaching

Teaching Fellow: Machine Learning for Economists	Spring 2026
Teaching Fellow: Machine Learning for Economists	Fall 2025

	Teaching Fellow: Environmental Economics and Policy	Summer 2025
	Teaching Fellow: Environmental Economics and Policy	Summer 2024
	Teaching Assistant Coordinator: Stata Lab	Spring 2024
	Teaching Fellow: Statistics	Summer 2023
Research Experience	*Research Assistant:* Richard Sweeney, Boston College	Summer 2021
	Research Assistant: Luigi Paciello, EIEF	Summer 2019
Programming and Software	Julia, Python, R, Stata, MATLAB, LaTeX, ArcGIS, Microsoft Excel	
Languages	Italian (native), English (fluent), French (basic)	
Other Interests	Climbing, hiking, skiing, photography, and cooking	
References	Richard Sweeney, Associate Professor, Boston College, sweeneri@bc.edu Michael Grubb, Associate Professor, Boston College, michael.grubb@bc.edu Edson Severnini, Associate Professor, Boston College, edson.severnini@bc.edu	